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ORIGINAL ARTICLE

First-year and final-year undergraduate students' perceptions of university mathematics departments

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Abstract
In many countries, concerns have been raised regarding the lack of participation of students in mathematics at the university level due to a dearth of skilled professionals to meet the needs of an increasingly technological, and thus mathematical, world. In this article, we report findings from a study in which we compared the experiences of first-year and final-year undergraduate students studying in the mathematics departments at two Australian universities. We focus on the participants' conceptions of the supports and challenges that they experienced in the mathematics departments, using a multi-modal data collection method, photovoice. We found key differences between first-year and final-year participants' perceptions of the learning environment when considering gender-related differences, such as an increasingly lower participation of women in mathematics and differential treatment by gender. Participants reported feeling supported by peers, at both year levels, and faculty members, particularly in first year. Participants from both year levels valued the collaborative learning environment of the classes and technological learning supports (e.g. lecture recordings) but raised concerns about the format and weighting of assessments. Although there were some year level differences, there were also commonalities to the participants' experiences, which indicates that some issues may be systemic. The knowledge that is gained from this research is crucial in understanding students' lived experiences and thus making suggestions to address university mathematics pipeline issues.

Keywords University-level mathematics · Pipeline issues · Gender issues · Student experience

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Introduction

Declining numbers of undergraduate students graduating in mathematics impact innovation in a world that is "becoming increasingly technological and significantly more mathematical" (Australian Academy of Science (AAS), 2016, p. 37). In Australia, a very small proportion (0.4%) of students enrolling in tertiary education plan to pursue degrees in the mathematical sciences (Australian Academy of Science, 2016). Consequently, several government and scientific organizations have stressed the need for increased participation in the mathematical sciences at the tertiary level (Australian Academy of Science, 2016; Australian Mathematical Sciences Institute [AMSI], 2017). Women remain a minority of students in university programs in the mathematical sciences, and women's proportion of the enrolments has been declining in recent years (Australian Academy of Science, 2016; AMSI, 2017; Johnston, 2015).

The current situation in undergraduate mathematics warrants further investigation to better understand the experiences that contribute to students' perseverance in the field. Hence, we conducted a study in which we focused on issues of student experience in mathematics degree programs, via a multimodal methodology, photovoice, combined with individual interviews. By providing students with an opportunity to discuss their experiences, both supportive and challenging, we have developed a comprehensive understanding of the issues that they face. We will use the insight that we have garnered to inform mathematics faculties and other stakeholders, via reports, workshops, and meetings, of ways to address students' concerns.

Broadly, the aim of this project was to understand the experiences of undergraduate students enrolled in mathematics degree programs, in order to explore how the mathematics departments support and/or challenge students. There is evidence that undergraduate students' experiences may be differentiated by gender (e.g. Ellis et al., 2016; Piatek-Jimenez, 2015), so questions were incorporated at the end of the individual interview protocol about students' perceptions of the role that gender plays in their experiences studying university mathematics. Although there was a general focus on gender in the study, here, we focus on how year level may play a role in students' experiences of studying mathematics (as a major or minor) at the undergraduate level. Our project was guided by the following research questions:

1. What are undergraduate mathematics students' experiences of university mathematics departments?

    a. What aspects of the departments do students find supportive?
    b. What aspects of the departments do students find challenging?
    c. How do students' experiences in the mathematics departments influence their career aspirations?
2. Are there differences in experiences by:

    a. Year level (first year versus final year)?
    b. Gender?
    c. Institution?

To address these questions, we utilized qualitative research methodologies, namely, comparative case study and photovoice. In this article, we focus on Research Question 2a. Specifically, to address this research question, we will focus on Research Questions 1a and 1b—the supports and challenges experienced by undergraduate students in their mathematics departments. In the next section, we discuss our theoretical framework, in order to provide an understanding of the stance from which the research was conceptualized.

Theoretical framework

This study is broadly framed by a feminist and social constructivist epistemological stance (e.g. Butler, 1999; Fosnot, 2005). We view knowledge as a human construction that is gendered, as well as being culturally, socially, and historically situated. Specifically, we conceive of disciplinary knowledge of mathematics, as well as the views of mathematics and mathematicians, as socially constructed, gendered, and linked to the specificities of time and place. With regard to the context of the study, we apply this lens to the students' experiences in mathematics degree programs by viewing their learning as "both a process of active individual construction and a process of enculturation into the mathematical practices of the wider society" (Cobb, 1994, p. 13).

Specific to the current study, we align with the sociocultural perspective outlined by Piatek-Jimenez (2015), wherein participation (or a lack thereof) in mathematics is conceptualized as being impacted by external barriers but ultimately reduced to the choices of individuals. We view the students in our study as having agency and electing to participate in mathematics at a non-mandatory level (i.e. university). We seek to understand what factors contribute to these choices, in particular the perceived supports and challenges within their undergraduate program, to explore what may be leading to the retention or attrition of other potential mathematics students at this stage of the mathematics pipeline. Working from these theoretical stances, in our study, we have been able to identify issues that should be considered by faculties to meet the needs of mathematics students.

We view gender and related concepts (e.g. sex) as non-binary social constructs (Butler, 1990, 1999; Fausto-Sterling, 2000; Ho & Mussap, 2019). In "gender issues" research in both mathematics education and other fields, sex and gender are often conflated, which is problematic since these constructs are not synonymous (Damarin & Erchick, 2010; Ho & Mussap, 2019). Namely, sex is a term referring to hormonal, physical, chromosomal, and other biological aspects of being female, male, or intersex (Fausto-Sterling, 2000). In contrast, gender is a term referring to the "behavioral, social, and psychological characteristics" (Pryzgoda & Chrisler, 2000, p. 554) of women, men, and non-binary individuals. Various terms are used, such as agender, pangender, and genderqueer, to refer to different variants of non-binary genders (i.e. genders outside the woman/man binary). Our focus is on gender, not sex, as we are interested in the gendered social roles and interactions of the participants. In contrast, in some previous research about gendered attrition from university mathematics, scholars have examined differences by sex rather than by gender. These studies were all done with a binary lens, which marginalizes non-binary students and excludes them from narratives developed by researchers. Our perspective and subsequent methodological choices allow non-

binary students to participate in our research without implicit marginalization by the language used in our research design. In this article, we are precise in our use of gendered language (women, men, and non-binary), as the "correct use of language is… extremely important" (Ho & Mussap, 2019) p. 1 in indicating one's theoretical stance.

Review of the literature

Tertiary mathematics education is an expanding field of research, and experts have suggested that research is needed about the students' experiences, as existing research often focuses on the teaching and learning of specific mathematical topics (Coupland et al., 2016). Although there are many studies (e.g. Hernandez-Martinez et al., 2011; Wade et al., 2017) about the transition from secondary school to university mathematics, there is a paucity of research about students' progressions throughout mathematics degree programs.

To provide context, we begin by providing a brief overview of studies focused on the pipeline into university-level mathematics. We then provide an overview of studies focused on general issues that affect students in university mathematics programs. We conclude by discussing studies specific to gender issues in university mathematics programs. Where possible, we have focused on studies conducted in Australia, in order to provide a better understanding of the educational context.

Pipeline into university mathematics

In Australia, no gender differences in participation in mathematics exist until the senior years of secondary school, when mathematics is no longer mandatory; importantly, there are no gender differences in achievement in mathematics in the pre-senior secondary years either (Watt et al., 2006). However, at the senior secondary school level, boys elect to study higher levels of mathematics than girls do, on average (Watt et al., 2006). For instance, at the Year 12 level in the Australian state of Victoria, higher proportions of boys than girls are enrolled in the two most advanced levels of mathematics, and this pattern has been consistent for more than 15 years (Forgasz & Leder, 2019). Various reasons contribute to gendered differences in participation, such as girls being picked on for liking mathematics (Piatek-Jimenez, 2015) and girls having lower mathematics self-concepts—the self-perception of mathematical ability compared with their peers—than boys, despite similar achievement (Githua & Mwangi, 2003; Hall, 2012; Sheldrake et al., 2015).

As mentioned, this disparity in senior secondary school mathematics participation increases the participation gap at the university level, as not participating in higher levels of mathematics in secondary school obstructs participation in mathematics at the undergraduate level (Correll, 2001). For girls (but not boys), participation in higher-level secondary school mathematics, especially calculus, is strongly linked to the choice of a mathematics or science major at a university (Trusty, 2002).

General issues in university mathematics

Researchers have identified many issues in the tertiary mathematics pipeline that have contributed to attrition, such as a lack of understanding of career pathways, poor

teaching, the demands of the course load, challenges transitioning to tertiary mathematics, loss of interest in content, and believing that another field of study will be more interesting (Fenwick-Sehl et al., 2009; Piatek-Jimenez, 2015; Ward-Penny et al., 2011; Wood & Solomonides, 2008). Piatek-Jimenez (2015) found that mathematically capable students choose other career options as they are uncertain about pathways beyond university. Concerningly, the undergraduate students in Ward-Penny et al.'s (2011) study felt that more focus was placed on memorization than understanding. Furthermore, marketing from universities focuses on research-level mathematics, despite only a fraction of graduates being likely to pursue that option (Fenwick-Sehl et al., 2009). Meaningful motivation to pursue undergraduate mathematics will require "careful statements about how, and to what extent, mathematics is truly used in various professions" (Fenwick-Sehl et al., 2009, p. 36; see also Wood & Solomonides, 2008).

Another challenge that students face, when beginning university studies in mathematics, is a change in the expectations for what is considered to be "legitimate mathematical activity" (Jablonka et al., 2017, p. 69), such as the requirements of a formal mathematical proof. Students often lack awareness of the differences between school and university mathematics, as well as the much higher mathematical requirements of the latter (Kosiol et al., 2018; Thoma & Nardi, 2018). A better understanding of the differences between high school and university mathematics could help to address attrition from university mathematics programs: If students have a better understanding of what to expect in university mathematics programs, they will be more likely to make an informed selection, based on what is a good fit for them.

Gender-specific issues in university mathematics

To meet the increased demand for mathematics graduates (Australian Academy of Science, 2016), there is a need to reduce the proportion of students who opt out at each level of mathematics education. Girls and women represent a large portion of potential mathematics students who are particularly vulnerable to leaving the field early, and much research attention has been placed on this issue (Fenwick-Sehl et al., 2009; Robnett, 2013).[1] The educational stage at which the gendered division in participation in mathematics is most significant varies between countries, but girls/women typically self-select out of mathematics as soon as it is no longer compulsory (Watt et al., 2006) or following the completion of undergraduate study (Piatek-Jimenez, 2015). A reported contributor to the gendered difference in participation at each successive level is cultural beliefs about gender and mathematics, which manifest as gender stereotypes and gender roles for the ways that people are expected to participate (or not) in mathematics (Correll, 2001; Sax et al., 2015; Simon et al., 2015).

Women enter into undergraduate mathematics having internalized these societal pressures and report a lower mathematical self-concept than ability-matched men (Sax et al., 2015; Watt et al., 2006). A positive mathematical self-concept is linked to both increased persistence and increased academic achievement (Linnenbrink-Garcia

[1] Although we conceptualize gender as a socially constructed spectrum, the studies that we report regarding gender issues in university-level mathematics all focused on women and men, rather than considering non-binary students. It is unclear how the gendered nature of mathematics impacts non-binary students, but, as another minority gender group in university mathematics, non-binary students may have some similar experiences to women students.

et al., 2018; Sax et al., 2015). This phenomenon holds a particular significance if we wish to lessen the attrition of students from mathematics, as "gender differences in the perceptions of task competence, over and above actual ability, foster gender differences in commitment to paths leading to that career" (Correll, 2001, p. 1692). Jouini et al. (2018) suggested that men are benefitted by gendered stereotypes about mathematics competency, which act as an "insulating layer that enables more self-confidence and more bold choices without fearing the impact of failure in terms of self-esteem loss" (p. 42). Although women are disproportionally affected, students of all genders and abilities appear to lose mathematical confidence as they progress through introductory calculus (Ellis et al., 2016). Sheldrake et al. (2015) suggested that the reasoning behind the students' self-beliefs and perception of ability is not always explored, so further research is needed to fully understand the development of these gendered differences.

Additional reasons for women's attrition from mathematics degree programs include a lack of support from faculty members, feelings of being invisible or not fitting in, a low proportion of women in the program, and a loss of interest in the subject area (Damarin, 2000; Herzig, 2004; Mastekaasa & Smeby, 2008; Rodd & Bartholomew, 2006). In contrast, protective aspects include social and academic support from friends and family, encouragement from educators, and personal characteristics such as determination and competitiveness (Gill, 2000; Hall, 2010; Robnett, 2013, 2016; Rodd & Bartholomew, 2006). The "leaky mathematics pipeline" for women in mathematics is a known issue worldwide (Watt et al., 2006) and so the need to compare the learning experiences of women across education levels is important.

Summary

The low enrolments in university mathematics programs follow from the decreased enrolment by students, especially girls, in mathematics at the senior secondary school level, once mathematics is no longer mandatory. Girls are less likely than boys to enrol in higher-level senior secondary mathematics courses, despite similar achievement in earlier courses. At the university level, students tend to struggle with the different, and higher, expectations for mathematics. With regard to gender, several issues have been found to be specific to women, such as a lower mathematical self-concept, feelings of not fitting in or being invisible, and issues regarding gender stereotypes and differential treatment. Although there is a great deal of research about the teaching and learning of university mathematics content, there is less research about the students' experiences in undergraduate degrees, particularly focusing on students of all genders and differences by year level.

Research design

The research project is a comparative case study of the mathematics departments at two Australian universities. We investigated the experiences of first-year and final-year undergraduate students, with a focus on gendered aspects of students' experiences. Experiences in the first year of university have been shown to be critical in supporting students, particularly women, to continue in the field (Herzig, 2004; King et al., 2015). The final undergraduate year is the time when students need to finalize decisions about

future careers and/or further studies. By having participants from different year levels, genders, and institutions, we were able to examine how these aspects may influence students' experiences. In the following sections, we provide an overview of the study's methodology, namely comparative case study and photovoice, and discuss the data sources, participants, and analysis methods.

Comparative case study

As a case study, our research involves "the study of an issue explored through one or more cases within a bounded system" (Creswell, 2007, p. 73). According to Stake's (1995, 2005) conception, our project is an instrumental case study, as we are focusing on a broader issue of which the case is representative, and a collective case study, as it is an instrumental case study extended to several cases (i.e. multiple case study design). The broader issue is the differential experiences and participation by gender and year level in studying university mathematics. To investigate this issue, we used a modified version of photovoice (Wang & Burris, 1997).

Photovoice

Photovoice involves participants taking photographs that are relevant to their lives in order to "promote critical dialogue and knowledge about important community issues through large and small group discussion of photographs" (Wang & Burris, 1997, p. 370), with the goal of reaching policymakers. The participants both create and discuss the data, increasing their autonomy in the research process, as they can "identify, define, and enhance their community according to their own specific concerns and priorities" (Wang & Burris, 1997, p. 374).

The use of photovoice has grown exponentially in the past few years, presumably due to the widespread use of smartphones. Several researchers (e.g. Cook & Quigley, 2013; Wilkinson et al., 2012) have used photovoice to learn about post-secondary students' experiences. However, we only know of three examples of photovoice in mathematics education research (Chao, 2012; Harkness & Stallworth, 2013; Tan & Lim, 2010), and these studies were not conducted at the post-secondary level.

We used a modified version of photovoice in which we began by conducting individual semi-structured interviews, focused on each participant's educational pathway into the mathematics degree program, experiences in the program, and career aspirations. Specifically, to provide background information, participants were asked about their initial interest in mathematics, secondary school experiences, decision to study mathematics at a university, and others' reactions to this decision. Most of the interview was focused on the participant's experiences at a university, such as their favourite and least favourite aspects of their program, supports available to them, desired changes to the program, gendered interactions, and future plans. After the interview, per the photovoice process (Wang & Burris, 1997), each participant took photographs to represent the supportive and challenging aspects (3–5 photographs per category) of the mathematics department. These photographs were submitted to the research team to be used as prompts in focus group interviews. Photographs can focus and encourage discussion in focus group interviews, as well as provide a different mode in which participants can express themselves (Whitfield & Meyer, 2005). In the focus

group interviews, we found that this was indeed the case, as the photographs directed the discussion and helped the students to contribute to the conversation. Rather than having a predetermined focus group interview protocol, the interview facilitator prompted each participant to explain each of their photographs. The participants took turns sharing their photographs and then the interview facilitator asked the other participants about their experiences with the challenge/support raised in the photograph. Finally, the interview facilitator guided the participants to discuss themes that they saw across the photographs.

Therefore, we have data from three sources—individual interviews, focus group interviews, and photographs—with the latter two intertwined. The individual interviews were audio-recorded, the focus group interviews were video-recorded, and the photographs were provided electronically to the researchers for further analysis. Multiple data sources allow for triangulation, "a process of using multiple perceptions to clarify meaning… [that] helps to identify different realities" (Stake, 2005, p. 454). Moreover, by comparing the data sources, "various strands of data are braided together to promote a greater understanding of the case" (Baxter & Jack, 2008, p. 554).

Participants

Data were collected in 2018 from two comparable, prestigious Australian universities (herein University X and University Y). Both have large mathematics departments, as shown in Fig. 1. In Australia, Bachelor of Science degrees, including mathematics degrees, are 3 years, full-time.

As shown, there is a drastic decrease in the EFTSL between first year and second year. Furthermore, women's proportion of the total enrolments decreases each year. Hence, these institutions are representative of the national enrolment trends mentioned earlier (Australian Academy of Science (AAS), 2016; AMSI, 2017), both overall and by gender. Data disaggregated by major or minor were not available, so we were not able to distinguish students who intend to complete a mathematics major or minor from

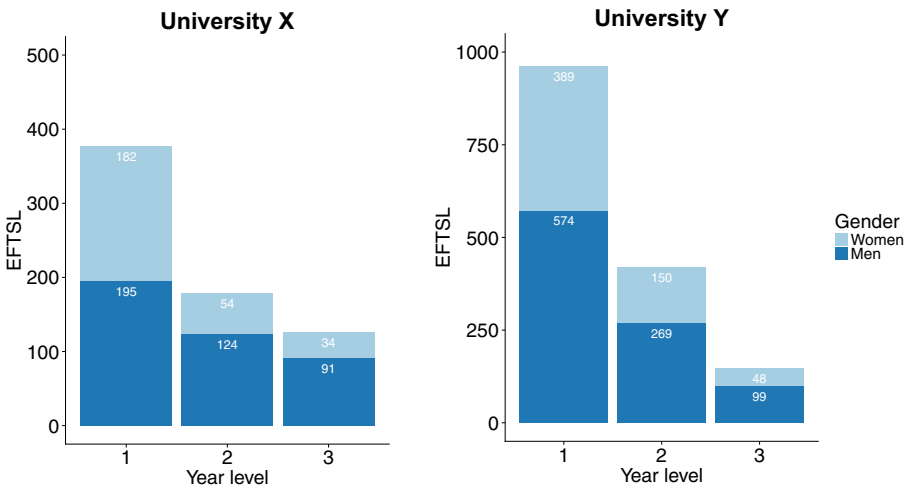

Fig. 1 Equivalent full-time student load (EFTSL) for University X (*left panel*) and University Y (*right panel*) enrolled in first-, second-, and third-year units

students who only need to complete mathematics units in first and second years as part of another program (e.g. science, business).

We recruited 26 participants for our study. Although most participants were traditional-aged undergraduate students (18–22 years old), five participants (19.2%) were mature-aged students. The participants have taken many different pathways into mathematics, such as transferring from other programs or working prior to beginning their studies. In Table 1, we provide information about the participants in the individual interviews by university, year level, and gender. As discussed, we view gender as a non-binary social construct and asked the participants to identify their gender in their own words during the interviews. All participants provided responses that aligned with binary categories, so the gender-related data are separated into "men" and "women" categories. On average, the first-year interviews were approximately 33-minutes long, whereas the final-year interviews were approximately 46-minutes long.

The focus groups comprised three students each (first-year or final-year students only in each focus group), as this group size was ideal in terms of providing space for all participants to share their photographs and contribute fully to the discussion. All participants who completed individual interviews were invited to take part in focus group interviews, but not all participants volunteered to do so. There were four focus group interviews held: one with first-year students and one with final-year students at University X, and two with final-year students at University Y. Therefore, the data about first-year students' experiences at University Y are strictly from individual interviews, since none of these participants volunteered for a focus group interview. On average, each final-year focus group interview was approximately 62-minutes long, whereas the first-year focus group interview was approximately 70-minutes long. Participant information is provided in Table 2.

In Table 3, the number of photovoice submissions by university and year level is shown.

There was a balance between "supportive" and "challenging" photovoice submissions across the entire dataset and by year level. However, the women provided far more photographs than did the men (average of 6.0 vs. 3.0 photos per person, respectively). Interestingly, 50% of the men stated that they had only prepared the photos within a day of their focus group interview. Similar gender differences in participation have also been found in other studies with undergraduate students (e.g. Deshler et al., 2019).

Table 1 Participant information: individual interviews

	First-year students			Final-year students		
	Men	Women	Total	Men	Women	Total
University X	3	5	8	3	3	6
University Y	4	2	6	2	4	6
Total	7	7	14	5	7	12

Table 2 Participant information: focus group interviews

	First-year students			Final-year students		
	Men	Women	Total	Men	Women	Total
University X	1	2	3	1	2	3
University Y	0	0	0	2	4	6
Total	1	2	3	3	6	9

Analysis

The multiple data sources and participant groups necessitated a complex and multistage approach to data analysis. First, the individual interview audio tapes and focus group interview video tapes were transcribed verbatim by a transcription company. These transcripts were read multiple times by all members of the research team to get an initial sense of the dataset. Then, the individual interviews were analyzed through a process of emergent coding (Bogdan & Biklen, 2007; Creswell, 2014). That is, after multiple readings of the transcripts to familiarize ourselves with the data, we created thematic codes based on themes that emerged from reading the transcripts. Some themes related to certain interview topics/questions, whereas others applied more holistically, across the interview. Due to the importance placed on the participants' explanations of the photographs, they were analyzed within the context of the focus group interviews, using similar emergent coding processes. We further analyzed the photographs using content analysis (Riffe et al., 2014), in order to provide additional detail and description that may not have been evident in the focus group interview videos. Specifically, we analyzed each photograph separately, examining its content and making links to the discussion from the focus group interview. We sorted the photographs into thematic categories. We then considered how these categories aligned with the categories that had emerged from the analysis of the individual interview data.

With regard to the comparative case study methodology, analyses occurred at multiple levels. To begin, thematic analyses of the entire dataset took place, in order to understand key themes regarding supports and challenges in the mathematics departments for all participants. Then, the findings were further considered with regard to the year level and gender of the participants, to see if there are any trends specific to

Table 3 Photovoice submissions from participants

	"Supportive" photographs			"Challenging" photographs		
	First-year students	Final-year students	Total	First-year students	Final-year students	Total
University X	6	9	15	7	5	12
University Y	N/A	16	16	N/A	16	16
Total	6	25	31	7	21	28

*Note: There was also one "mixed" photograph provided by a final-year student at University Y. Hence, the total dataset comprised 60 photographs

these groups. Each institution was considered separately to develop a holistic understanding of each mathematics department.

Findings

In the following sections, we present findings from all aspects of the project—individual interviews, focus group interviews, and photographs—focusing on comparisons by year level. We begin by discussing findings that pertain to the first-year participants only, followed by findings that pertain to the final-year participants only. We conclude by describing findings common to participants from both year levels.

Findings specific to first-year students

Generally, the first-year participants spoke very positively about their experiences at a university. Participants at both institutions reported feeling supported by faculty members and generally not experiencing any gender issues. Issues regarding mathematical software were raised by participants at University X.

Faculty support

The first-year participants reported a positive experience overall and found the structure of the university learning environment to be valuable. They described the faculty members as passionate and helpful. The participants attributed any lack of achievement in mathematics to the challenge of transitioning between high school and university, rather than any distinct factors in the mathematics departments. The participants explained that if they had any difficulty adjusting, they felt supported and encouraged by the faculty members to continue participating in mathematics. For instance, Han (first year, man, University Y, individual interview) reported: "They've been very patient to me, even though sometimes they explain the method of tackling question to me once, but I'm still bit confused and yeah, they just repeat the process until I fully understand the question". Other participants noted the availability of faculty members for support through office hours.

Lack of gender issues

In terms of gender-related issues, first-year participants reported an even gender distribution of students in their classes. However, as noted earlier, this gender balance is likely due to the requirement of first-year mathematics classes for many programs outside of mathematics. Encouragingly, the first-year participants noted minimal perception of personal differential treatment by gender (7.1%; one woman) or of others being differentially treated by gender (7.1%; one woman).

Mathematical software issues

First-year students at University X raised concerns about the lack of guidance and support provided when they were introduced to Mathematica. They were required to

complete an assignment using Mathematica and felt that they had not been given sufficient instruction on the software, as discussed by two participants in Focus Group #1 (first-year students, University X):

Chen: For Mathematica, like I found it a bit hard for the first assignment to use it. I could not – like, I had to keep Googling like, "What do I do?"

[…]

Fiona: It was, "By the way, Questions #2 and #3 require Mathematica" and it's like, "What's Mathematica?" "Make sure to download Mathematica". Yeah, we were so lost. [Fiona put her head in her hand at this point] So, after that and making sure you download Mathematica, then there was a 10-min crash course of the most simplest [*sic*] things.

As a result of the lack of instruction on the software, the participants felt disenfranchised with the assessment, as they believed that it extended beyond an appropriate level of challenge.

Summary of findings specific to first-year students

Generally, the first-year participants were very positive about the experiences in their mathematics programs. They found the faculty members to be supportive and typically had not experienced or seen differential treatment by gender. The only issue raised (at University X) was a lack of instruction regarding mathematical software. As noted, the interviews with the first-year participants were shorter than those with the final-year participants (33 vs. 46 minutes, respectively); since the former group was newer to the environment, they had fewer experiences to discuss. Furthermore, many issues that the first-year participants raised (e.g. regarding assignments) were also raised by the final-year students, which will be discussed in the final section of the findings.

Findings specific to final-year students

The final-year participants were more critical than the first-year students of the support provided by the faculty members, as well as other aspects of their experiences in the mathematics departments. We begin by discussing this issue and then address the final-year participants' critiques of the misalignment between pedagogy and assessment. Finally, we report concerning gender issues raised by the final-year participants, particularly the women.

Critiques of faculty members

The final-year participants generally felt supported by the faculty members but noted that the quality of support varied widely by faculty member. Although the final-year participants were accepting of the university teaching structure and understanding its rationale, they were less enamoured than the first-year participants. For instance, Lawrence (final year, man, University Y, Focus Group #4) remarked: "I don't like big lectures. I'm happy to have small lectures or more workshop-style learning, but there's a simple reality of the finances to this university, I get that. We happen to have lectures just on the financial side of things".

Some final-year participants commented on faculty members' lack of expertise in pedagogy. For instance, Beatrice (final year, woman, University Y, individual interview) reported:

Other things that have been strugglesome about maths, disengaged tutors or lecturers, people who aren't as committed to educating. When you are a student who's there to learn, having someone who's not willing to teach is so challenging, or, I guess, lecturers who aren't education trained in any way. You go to someone, "I don't understand this." And they explain it in a complicated way. And you say, "No, but I don't understand this thing." They have no framework for how to simplify it.

Some participants suggested that the faculty members' lacking pedagogical skills were due to a focus on research rather than on teaching. Such issues with pedagogy were particularly noted by students who were completing double degrees in mathematics and education, as they had already developed an understanding of supportive pedagogical practices.

Another final-year student, Eloise (woman, University Y, individual interview), discussed her conception of an ideal response from faculty members when students are struggling:

A big thing is when a student fails a subject, have someone to catch them almost. And I know that's not how a university works, but just 15 minutes with the lecturer and the lecturer can be like 'What are you doing?' like 'Why did you fail?' or 'Let's have a look at your exam and let's figure out where you missed up', things like that.

Eloise continued by describing the impact that a faculty member had on her by supporting her after she had failed a subject. Eloise noted that she was more likely to attend this class since she knew that the faculty member would be watching for her; hence, she did better due to her increased attendance.

Generally, final-year students were understanding of the resource restraints, but there was a desire for more intimate learning environments and exchanges with lecturers. Final-year students were very attentive to the implicit educational values of staff as demonstrated in minor interactions, such as providing email addresses and noting their availability for support.

Misalignment between pedagogy and assessment

Final-year participants raised issues regarding the individual nature of assignments. Namely, they noted inconsistent messaging between the value placed on working collaboratively in class and the individual nature of assignments. Researchers consistently argue that pedagogical and assessment practices need to be aligned (e.g. Secolsky & Denison, 2011; Suurtamm et al., 2016), but this did not seem to be common in either institution. Particular concerns were raised regarding the messaging surrounding collusion:

When you are in first year, it's very restrictive having this thing [collusion policy] and […] you are not allowed to plagiarize, but when I first saw that, that told me that I then wasn't allowed to work with anybody and isolated my learning. (Kristin, final year, woman, University X, Focus Group #2)

Some students had received verbal clarification from lecturers about distinctions between collaboration and collusion, but this was not a consistent experience for all participants. Hence, due to the lack of clarity, students tended to act in a cautious

manner when working on assignments. For instance, Lawrence (final year, man, University Y, individual interview) reported that he and his friends "basically just work on the basis of work together and write up separately. Because we all find that how we write it up diverges enough that that's an acceptable level". Although this strategy may be functional, it would be better if clarity was provided regarding this issue.

Gender issues

Final-year students, especially women, were acutely aware of the declining participation of women in mathematics as they progressed through their program. Beatrice (final year, woman, University X, individual interview) noted: "In the early years when it's compulsory, there's a pretty good gender balance, and as you go further and further up, there are less girls, I've noticed in the tutes". Although most final-year participants were able to provide a clear estimate of the gendered ratio of their classmates (approximately one-quarter to one-third women by their final year of studies), women were hyper-aware of a decreased proportion of women in higher year classes (Fig. 2). Notably, some of the women had counted the exact numbers of men and women in their classes, whereas the only participants who were unsure of the ratio were the men.

In final year, compared to first year, there was an increase in the perception of personal differential treatment by gender (16.7%; two women) and of others being differentially treated by gender (41.7%; three women, two men). However, beyond these clear "yes" responses, three more participants were unsure whether they had personally been treated differentially by gender (25.0%; two women, one man). For example, two final-year women participants from University X shared similar stories of a classmate (a man) taking a marker out of their hand while they were working on a mathematics problem on the whiteboard during class. One participant interpreted this action as gendered, whereas the other said: "I don't think it's because I'm female and there was a perceived gender difference. I think they were just twats." (Beatrice, final year, woman, University X).

Lawrence (final year, man, University Y, individual interview) shared the following story of differential treatment by gender of one of his friends:

She had one instance where someone once said something along the lines of – basically she'd got a good exam mark and they'd be like, "Oh yeah, but they mark you easy [*sic*] because you're a girl," at which point literally seven other guys jumped down his throat. […] She was like, "Before I had a chance to respond, literally every other guy in that group just tore this guy to shreds, and he never, literally never, turned up to another lecture."

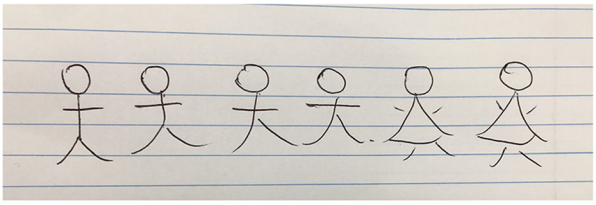

Fig. 2 Photograph provided as evidence of "challenging" aspects of the mathematics department from Eloise (final year, woman, University Y, Focus Group #3)

This story provides an example of a quandary: Were the men acting as allies or "saviours" for their woman classmate? It is interesting that the men reportedly spoke up against their sexist classmate before the woman had a chance to respond. Without being present for this interaction, or the woman's comments about it, it is hard to assess the nature of the event. Certainly, though, it is a positive ally behaviour that several men challenged the sexist man on his comments (Association for Women in Science, n.d.).

There was a general trend for the final-year women to internalize the pressure that they feel to achieve in mathematics, as well as fear that impeded their engagement. For instance, Eloise (final year, woman, University Y, individual interview) shared the following story:

Someone once called me Barbie once in class. They're like, "You look like a Barbie studying maths" and I was like, "What?" […] That always stuck with me that I did not want to look like this stupid bimbo in the class. And so sometimes I'll be like, "Do not talk." Even if I'm burning to ask a question, I'm like, "I'll look like an idiot. I don't know how to do this but I'll look like an idiot".

Such constraints arguably lead to lower understanding, as misconceptions and misunderstandings are not cleared up in class. The final-year women participants reported feeling significant internal pressure to justify their own continued participation and felt that any personal failure would contribute to gender inequity. Concerningly, Beatrice (final year, woman, University X, individual interview) reported that, "Failing at a concept often feels like failing as a girl. Or, as a female in mathematics, I feel very representative of that." Similar to Eloise, Irina (final year, woman, University X, individual interview) suggested that not only her gender, but also her appearance, may have led to differential treatment:

I do not know, even that I'm blonde and female, it's like very – last picked sort of scenarios happen quite a bit or just assuming – even if I would go to write on the board and I'd halfway get through something and then someone else would swoop in and be like, "Actually, I know better" but they are writing the same thing as me.

It is troubling that so many final-year women raised issues regarding their own mathematical identities and self-concepts, which were often related to the interactions with their peers in their mathematics program.

Summary of findings specific to final-year students

In comparison to the first-year participants, the final-year participants raised more concerns about their experiences in their mathematics programs. In particular, the final-year participants highlighted a lack of pedagogical skill shown by some faculty members, whereas others were described as very supportive. The final-year participants also critiqued the lack of alignment between the collaborative working environment in classes and the individual nature of assessments, particularly surrounding fear and uncertainty linked to collusion policies. Many troubling gender issues were raised by the final-year students, and internalized feelings of incompetence, lacking confidence, and/or a lack of belonging raised by the final-year women.

Findings common to students from both year levels

Many issues—both supports and challenges—were raised by participants from both year levels. Specifically, participants discussed the physical environment, collaborative

learning, assessment weighting, lecture recordings, and the gendered composition of their lecturers. Unless otherwise noted, the issues raised were consistent at both institutions.

Physical environment

Participants from University X and University Y had very different impressions of the physical environment in their respective mathematics department building. Although the mathematics department buildings at both institutions are older structures, renovations have occurred to both recently, with varying levels of success.

At University X, participants consistently commented positively about the fact that all of the surfaces (e.g. walls, tables) can be written on using whiteboard markers (see Fig. 3), which promoted collaboration and risk-taking. For instance, Kristin (final year, woman, University X, Focus Group #2) commented: "I think it helps push out an idea that you try to work on and save the environment and things. I like that the building was designed that way".

Furthermore, some participants noted that the room design allowed lecturers to present the mathematical content in an active, cohesive manner, which promoted deeper understanding of the mathematical content as well as the process of doing mathematics: "They'll start writing, and by the end of it, the whole wall is filled with math and you consistently just look back" (Max, final year, man, University X, individual interview).

Students from University Y had heard about the classrooms at University X and were quite envious that their institution did not have the same types of classrooms. Although their departmental building had recently been renovated, the participants from University Y noted that the furniture was not functional and that there was

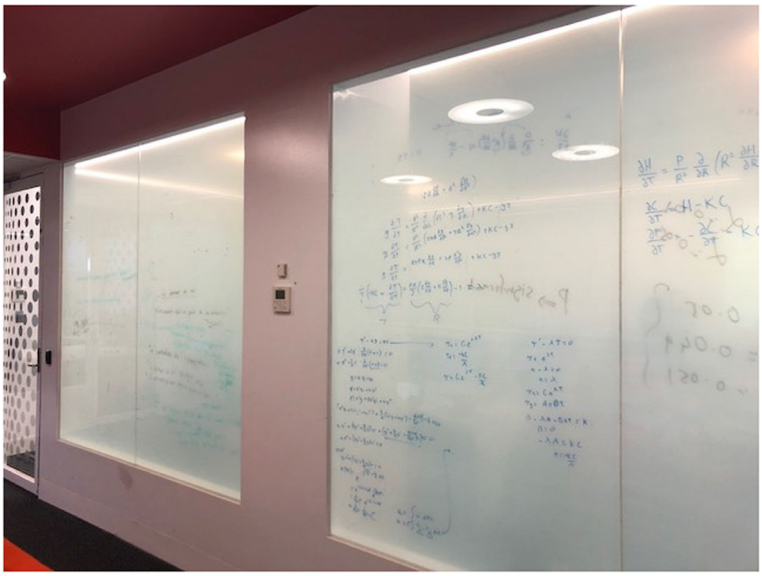

Fig. 3 Photograph of windows with mathematical writing provided as evidence of "supportive" aspects of the mathematics department (Kristin, final year, woman, University X, Focus Group #2)

insufficient space for collaboration, as noted in the following exchange from Focus Group #4 (final-year students, University Y):

Eloise: Okay. This [lecture] theatre has just been redone, except that the tables are too low and the chairs are too high.

Daria: It was like, yes, they have redone it, but it's not functional.

Eloise: It's not functional, yeah.

Daria: Yeah. That was my photo. They put in some effort, but it wasn't—it's not functional.

[…]

Daria: I walked in after the break and I was like, "Oh, wow!" New chairs, new tables, how exciting. Then sat down and I was like, "Oh, yikes".

[…]

Fletcher: Yeah. Some of the classrooms aren't laid out well for what they are supposed to be used for. [Room number] is a tutorial room, and all the desks are lined up in rows.

Thus, the participants showed an awareness of the lack of alignment between the physical environment and the pedagogical purposes of the rooms.

Concerningly, issues of accessibility were raised by participants at University Y. They noted that the multi-storey mathematics department building is not accessible, as it does not have an elevator or accessible bathrooms. Additionally, Eloise (final year, woman, University Y, Focus Group #3) shared that some of the large lecture halls do not have adaptive supports for students with hearing issues. She noted that her friend who wears hearing aids "struggled so hard, because she couldn't hear what was going on. The lecture cam[era] was good, but […] Not when you don't have the information […] So that was again lecture theatres and accessibility, that ties in together".

Beyond the lack of provisions for students who require adaptive supports, the women reported that the bathrooms were unsanitary (e.g. overflowing sanitary bins) and that there were always line-ups to use them. In Focus Group #3, Eloise (final year, woman, University Y) reported that "sometimes it's quicker to trek halfway across campus" to use a bathroom in another building rather than wait to use the women's bathrooms in the mathematics department building, since "queuing up is so bad" (Daria, final year, woman, University Y). The men from University Y confirmed that there were no such issues with the men's bathrooms in the mathematics department building.

Collaborative learning

In conjunction with the design of the learning environment, namely University X, participants reported that the tutorials are interactive, focusing on collaboration and collective understanding. For many of the international first-year participants, working in this way was new. However, as discussed by Chen (first year, woman, University X, Focus Group #1), students saw the value in learning in this manner: "This helped me to – I don't know, like the subject better and also just see how other people – be able to, like, combine and collaborate I think to find the solution, which was really fun". The collaborative style of learning allowed the first-year participants to meet other students and to develop a better understanding of the content. Erwin (first year, man, University X) provided the photo in Fig. 4 as an evidence of the collaborative learning environment.

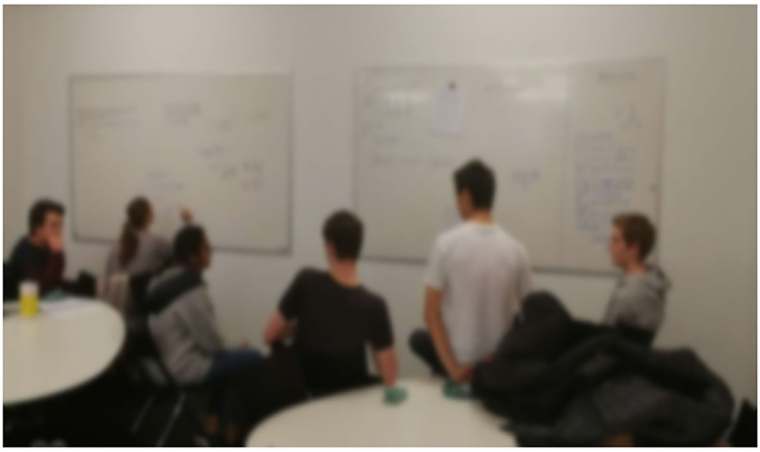

Fig. 4 Photograph provided as an evidence of collaboration in classes of the mathematics department (Erwin, 1st year, man, University X, Focus Group #1)

Almost all participants reported that peers were a major support in their academic success and persistence: "Subjects where I've got friends, I do better than in subjects when I don't […] You feel like you're the only person who doesn't get this, but then it's like probably half the cohort also doesn't" (Eloise, final year, woman, University Y, Focus Group #3). Although some peer support developed from relationships nurtured within the collaborative class environment, other support was provided by peers who were not enrolled in the same classes as the participants, such as friends from other institutions or from their university residence. Han (first year, man, University Y, individual interview) described his support networks that developed from class as follows:

We have peer support both in class and after class discussions towards questions. In tutorial class, we do the questions together. If I get stuck or I simply make mistakes when I go through the questions, my peers would mark me or will correct me in some ways. After the tutorial classes, I do have some friends that I have class with, so we go to lectures together and sometimes we do the questions after class from the question booklet. We run a bit of mock exam sessions by doing past exam papers together, then we cross marking [sic], we correct each other's mistakes and yeah.

In contrast, Andrew (first year, man, University Y, individual interview) reported that peers from his university residence assisted him in developing conceptual understanding of mathematical content that was only taught procedurally in one of his classes.

Weighting of assessments

Participants were concerned about the weighting of marks for assignments and exams. The disproportionate amount of time required for assignments compared with other means of assessment (e.g. exams) was an issue that was commonly raised. Participants reported that assignments would take a whole weekend to complete but would be worth a tiny proportion of their final grade. Irina (final year, woman, University X, individual interview) noted that "they heavily weight the exams and then the assignments which might take you eight or 12 hours to do and will only be like 3% or something like that.

That has been quite frustrating". Such experiences were described as "degrading" and "demotivating", and participants expressed a desire to see a reduced time commitment for completing assignments. Similarly, participants suggested that "lecturers need to find better ways of gauging how long things will take students" (Griffin, final year, man, University X, Focus Group #2).

Not only was the disproportionate amount of time spent on minor assignments frustrating to students, they also felt that too much of the final grade was allotted to final exams. Participants raised concerns about students who struggled in that type of high-stakes assessment situation: "The exam being 80%, it's rough for a lot of people. I feel I do well in exams so it's fine for me, but I feel bad [*sic*] for all the people that are really smart and then just don't perform in exams" (Jasmine, final year, woman, University Y, individual interview). Generally, participants understood that individual assessments, such as exams, were required as evidence of each individual's understanding. Nevertheless, many participants expressed a desire for less heavily weighted exams, or at least multiple exams rather than a single final exam.

Lecture recordings

Students at both institutions valued classes that had lecture recordings, as these tools supported their learning in many ways. Participants commented that they were rarely able to fully understand content by attending lectures in person and found that the ability to pause and rewind the recordings mitigated these issues. These capabilities were particularly valued by participants who described themselves as "slow learners" or "not that good at maths". Therefore, lecture recording (Fig. 5) enabled students to self-pace and avoid the potential for judgment when asking for clarification on content.

Furthermore, lecture recordings provided equitable access to content for students who have other commitments such as a long commute time or part-time work.

Despite these resources being overwhelmingly viewed as a useful asset to learning, lecture recording was not as common at University Y as University X. Participants

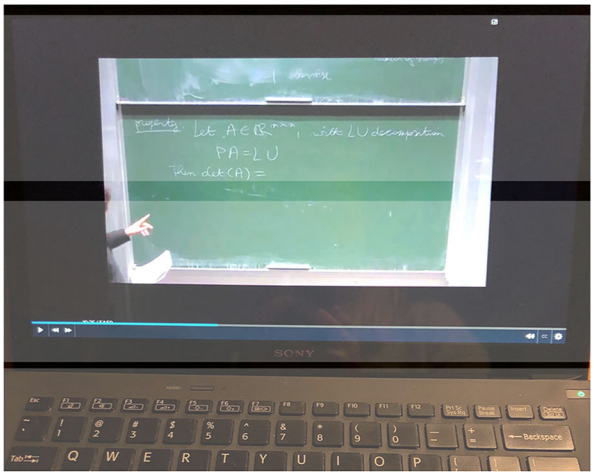

Fig. 5 Photograph of lecture recording provided as evidence of "supportive" aspects of the mathematics department (Irina, final year, woman, University X, Focus Group #2)

from University Y reported that some lecturers refused to record lectures, as they thought that would discourage attendance. Even when students requested lecture recordings to assist with their learning, some faculty members refused to accommodate this request, saying, "I've been teaching for 20 years and this is how I always do it, so I'm not changing" (Jasmine, final year, woman, University Y, individual interview).

Gender imbalance—lecturers

At both year levels, at both universities, participants reported that they had very few women lecturers. However, there were stark differences between institutions in terms of having been taught mathematics at a university by a woman. At University X, most participants (10 of 14; 71.4%) reported that they had never had a woman lecturer. Two participants reported having had one woman lecturer, whereas the remaining two participants were not clear about their experiences (e.g. "mostly guys"—Ahmed, 1st year, man, individual interview). In contrast, at University Y, most participants (8 of 12; 66.7%) reported having at least one woman lecturer. Although some participants from University Y had only been taught by one woman lecturer, even by the final year of their studies, as Katie (final year, woman, University Y, individual interview) shared, seeing women in prominent positions of power, both within the mathematics department and the faculty of science more broadly, helped mitigate the lack of direct role models in classes. Role modelling was recognized as important by the final-year women participants in particular: "You just don't realize, like, if there's no role modelling, it's very hard to picture yourself doing something" (Eloise, final year, woman, University Y, individual interview).

Summary of findings common to both year levels

Issues pertaining to the physical environment were commonly raised at both institutions, with students from University X commenting positively on the building's design and students from University Y noting concerns regarding accessibility. At both institutions, participants reported that classes focused on collaboration, which deepened their understanding and helped them to make connections with peers. Participants from both institutions were concerned about the weighting of assessments, with a large proportion of the grades allocated to a single exam and only a small proportion allocated to assignments that were time-intensive to complete. Lecture recordings were valued by the participants, but there was some resistance to this technology by lecturers at University Y. Finally, at both institutions, few of the lecturers are women, so students have had little opportunity to be taught by women, particularly at University X.

Concluding remarks

In this study, we investigated the supports and challenges experienced by undergraduate mathematics students at two Australian universities. For the purposes of this article, we focused on year-level differences, as well as gender issues, which are a general focus of the study. We found that the first-year students' experiences in the mathematics departments were quite positive and that these students feel supported by

and connected to their mathematics departments. In contrast, the final-year students had more critiques of their experiences in their mathematics departments, although they did recognize institutional constraints. As students progressed through their undergraduate studies in mathematics, there was evidence of increasingly gendered experiences, including disparaging interactions with faculty and, particularly, peers. Such experiences, combined with a broader societal context in which women's mathematics abilities are often denigrated, led to many of the final-year women participants struggling with mathematics self-concept. Such outcomes raise the question of whether we should be considering the experiences of these women a "success" in terms of retention of women in mathematics (see Hall & Suurtamm, 2018).

The interactive, collaborative nature of the mathematics classes at both universities—particularly supported by the physical learning environment at University X—provides evidence of the instructors' understanding of quality teaching practices, such as interaction and cognitive activity of learners (Bergsten, 2007; Laursen & Rasmussen, 2019). Collaborative working environments in mathematics classes have also been found to support women in particular to overcome affective barriers, such as low self-confidence (Steele et al., 2005), and to help students to develop transferable skills like teamwork and critical thinking (Laursen & Rasmussen, 2019).

However, the final-year participants noted inconsistent messaging between the value placed on working collaboratively in class and the individual nature of assignments. Researchers consistently argue that pedagogical and assessment practices need to align (e.g. Secolsky & Denison, 2011; Suurtamm et al., 2016). Hence, changes to assessment practices, such as including group assignments, would address students' concerns and would be consistent with best practices in assessment. Students at both year levels also raised concerns about the weighting of assignments, particularly the large proportion of the final grade that was allotted with a single exam. Decreasing the weighting of exams, as well as introducing alternative assessment practices, may decrease the anxiety that some students feel during high-stakes examinations (Deshler et al., 2019).

It is important to address not only structural issues in mathematics programs but also more subtle threats like microaggressions. None of the women in the study reported experiences of hostile sexism, but several reported experiencing microaggressions, "brief, everyday exchanges that send denigrating messages to certain individuals because of their group membership (e.g., people of color, women, or LGBTs [lesbian, gay, bisexual, transgender])" (Sue, 2010, p. 24). Gendered microaggressions are often experienced by women in the fields that remain dominated by men and that are perceived as being masculine, such as science, technology, engineering, and mathematics (STEM) fields (Barthelemy et al., 2016; Yang & Carroll, 2018). In our study, some examples of gendered microaggressions were unclean women's (but not men's) bathroom facilities and help offered to a woman who was doing the same mathematical work as a man; according to Sue and Capodilupo's (2008) and Nadal's (2010) taxonomies of gendered microaggressions, these examples would be considered, respectively, an "environmental" microaggression and an "assumption of inferiority" microaggression. Researchers suggest that microaggressions can be addressed through educational and awareness training for both faculty members and students, as well as immediate responses to such gendered microaggressions when they occur (Barthelemy et al., 2016; Yang & Carroll, 2018).

Women lecturers act as mentors and role models, and therefore play an important role in helping women to continue in mathematics studies at the university level (Bettinger & Long, 2005; Herrmann et al., 2016; Steele et al., 2005). At both University X and Y, there is a dearth of women mathematics faculty members, and many participants had never been taught university-level mathematics by a woman. Although increasing the number of women faculty members is a valuable longer-term goal, a more immediate solution, which has mentoring and role modelling benefits (Steele et al., 2005), is to provide women guest speakers to discuss their experiences as women in mathematics and to demonstrate possible career paths. Another solution is to provide peer mentoring to women students, particularly by other women students. Such programs have been successful in increasing women's sense of belonging, developing a supportive community, and consequently decreasing attrition from STEM fields (Bryant & Alvarado, 2017; Dennehy & Dasgupta, 2017).

In addition to the findings reported in this article, we provided an example of the use of photovoice, a novel method of data collection in mathematics education, particularly at the university level. By sharing our experiences, we may assist colleagues in expanding their methodological repertoires. Photovoice allows participants to express their feelings about their experiences in a democratic, participatory manner. Visual representations allow participants to share their ideas in a mode that may be more accessible, thus providing unique insights into their experiences. Furthermore, the use of photographs can lead to insights into other aspects of the students' learning environments. For instance, although not the focus of the issues raised by participants with their photographs, during analysis, we noticed that the mathematics shown was overwhelmingly presented in an equation format, with little visual or physical representation shown. Researchers (e.g. Goldin, 2002; Wood et al., 2007) have shown that using multiple representations of mathematics supports a wide range of students engaging with and understanding mathematical concepts. Without the photographic evidence of the students' experiences, such a finding would have been missed.

The use of a multimodal methodology, photovoice, combined with individual interviews provided us with important insights into undergraduate mathematics students' lived experiences. The themes that emerged from our analysis reflected concerns previously identified in the extant literature, as well as hidden issues faced by these students. By better understanding the complexity of students' experiences, we can begin to envision solutions to the problems raised. Hence, our findings will inform practice at the participating universities and hopefully increase retention of mathematics majors, thus addressing the National Innovation and Science Agenda aim to boost the number of Australian graduates in science and mathematics as a strategy to build Australian capabilities for innovative economies (Australian Government Department of Industry, Innovation and Science, 2015). Retention and, more importantly, rewarding experiences in university mathematics programs are only possible when all students feel that the mathematics departments are places where they belong.

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